

```
In [1]: df <- read.csv("https://raw.githubusercontent.com/blacktreeM/econ/refs/heads/main/cbb.csv")
```

```
In [3]: colnames(df)
```

```
'TEAM' · 'CONF' · 'G' · 'W' · 'ADJOE' · 'ADJDE' · 'BARTHAG' · 'EFG_O' · 'EFG_D' · 'TOR' · 'TORD' · 'ORB' · 'DRB' · 'FTR' · 'FTRD' · 'X2P_O' ·  
'X2P_D' · 'X3P_O' · 'X3P_D' · 'ADJ_T' · 'WAB' · 'POSTSEASON' · 'SEED' · 'YEAR'
```

```
In [2]: model <- lm(W ~ EFG_O + TOR + ORB + FTR, df)  
summary(model)  
coeffs <- abs(coef(model)[-1])  
coeffs / sum(coeffs) * 100
```

Call:

```
lm(formula = W ~ EFG_O + TOR + ORB + FTR, data = df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-13.3986	-2.9140	-0.0847	2.8572	14.2176

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-45.06306	1.50256	-29.99	<2e-16 ***
EFG_O	1.12558	0.02201	51.15	<2e-16 ***
TOR	-0.98110	0.03183	-30.82	<2e-16 ***
ORB	0.62216	0.01587	39.21	<2e-16 ***
FTR	0.14057	0.01235	11.38	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.077 on 4244 degrees of freedom

Multiple R-squared: 0.6066, Adjusted R-squared: 0.6062

F-statistic: 1636 on 4 and 4244 DF, p-value: < 2.2e-16

EFG_O: 39.226944890251 **TOR:** 34.1915558646277 **ORB:** 21.6825102448299 **FTR:** 4.89898900029139

Task: Your boss noted that the relative importance of the Four Factors may vary by season. She wants a visualization to compare the magnitude of these effects over time. Create a line graph showing the regression coefficients for each factor by year.

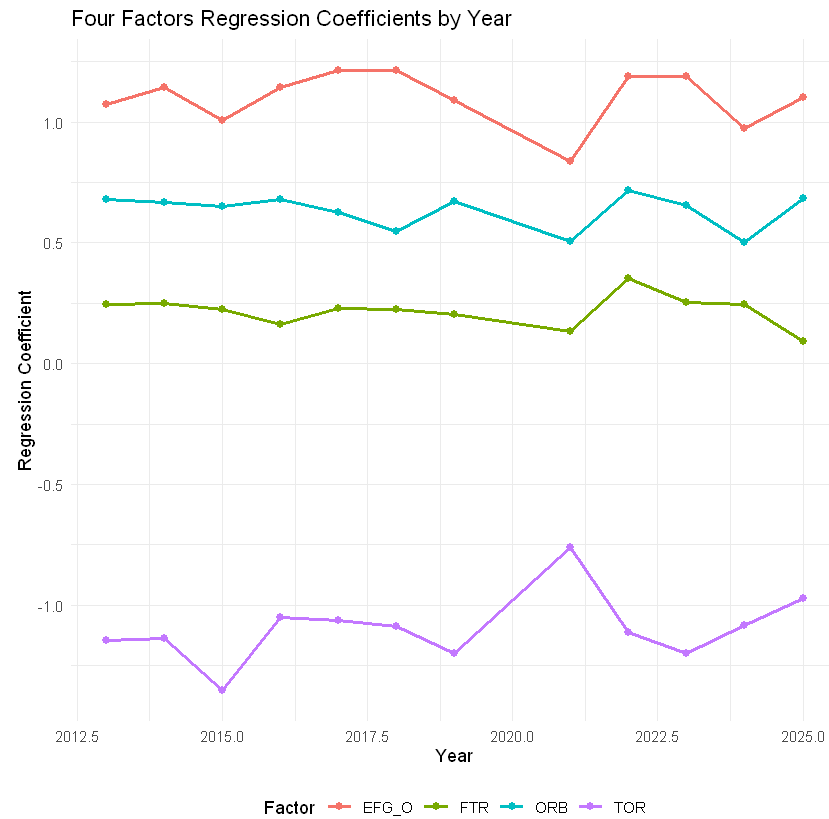
```
In [4]: library(tidyverse)
results <- df %>%
  group_by(YEAR) %>%
  group_modify(~ {
    model <- lm(W ~ EFG_O + TOR + ORB + FTR, data = .x)
    coef(model)[-1] %>%
      enframe(name = "Factor", value = "Coefficient")
  }) %>%
  ungroup()

ggplot(results, aes(x = YEAR, y = Coefficient, color = Factor, group = Factor)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Four Factors Regression Coefficients by Year",
    x = "Year",
    y = "Regression Coefficient",
    color = "Factor"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
```

```

Warning message:
"package 'tidyverse' was built under R version 4.5.2"
Warning message:
"package 'ggplot2' was built under R version 4.5.2"
Warning message:
"package 'tidyr' was built under R version 4.5.2"
Warning message:
"package 'dplyr' was built under R version 4.5.2"
Warning message:
"package 'stringr' was built under R version 4.5.2"
— Attaching core tidyverse packages — tidyverse 2.0.0
—
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats   1.0.1    ✓ stringr    1.6.0
✓ ggplot2   4.0.2    ✓ tibble     3.3.0
✓ lubridate 1.9.4    ✓ tidyr      1.3.2
✓ purrr     1.1.0
— Conflicts — tidyverse_conflicts()
—
✗ dplyr::filter() masks stats::filter()
✗ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```



In []: